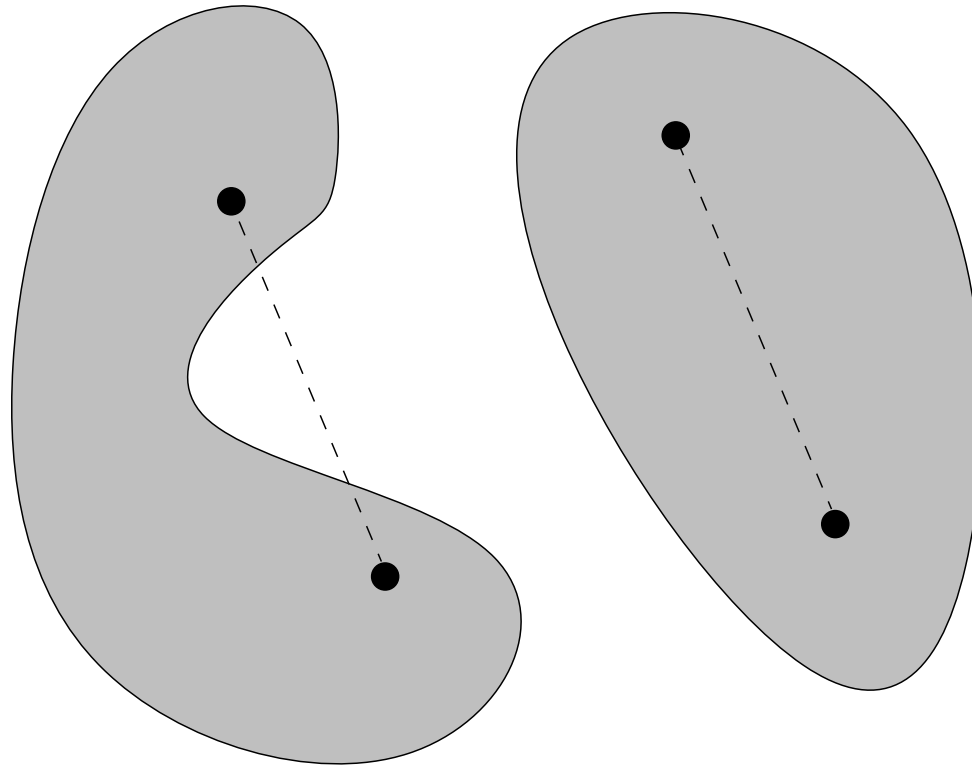


Convex sets



Convex sets

Def. A set $C \subset \mathbb{R}^n$ is called **convex**, if for any $x, y \in C$, the line-segment $\theta x + (1 - \theta)y$ ($0 \leq \theta \leq 1$) also lies in C .

Combinations

- ▶ **Convex:** $\theta_1 x + \theta_2 y \in C$, where $\theta_1, \theta_2 \geq 0$ and $\theta_1 + \theta_2 = 1$.
- ▶ **Linear:** if restrictions on θ_1, θ_2 are dropped
- ▶ **Conic:** if restriction $\theta_1 + \theta_2 = 1$ is dropped

Convex sets

Theorem (Intersection).

Let C_1, C_2 be convex sets. Then, $C_1 \cap C_2$ is also convex.

Proof. If $C_1 \cap C_2 = \emptyset$, then true vacuously.

Let $x, y \in C_1 \cap C_2$. Then, $x, y \in C_1$ and $x, y \in C_2$.

But C_1, C_2 are convex, hence $\theta x + (1 - \theta)y \in C_1$, and also in C_2 .

Thus, $\theta x + (1 - \theta)y \in C_1 \cap C_2$.

Inductively follows that $\bigcap_{i=1}^m C_i$ is also convex.

Convex sets – more examples

♡ Let $x_1, x_2, \dots, x_m \in \mathbb{R}^n$. Their **convex hull** is

$$\text{co}(x_1, \dots, x_m) := \left\{ \sum_i \theta_i x_i \mid \theta_i \geq 0, \sum_i \theta_i = 1 \right\}.$$

♡ Let $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$. The set $\{x \mid Ax = b\}$ is convex (it is an **affine space** over subspace of solutions of $Ax = 0$).

♡ **halfspace** $\{x \mid a^T x \leq b\}$.

♡ **polyhedron** $\{x \mid Ax \leq b, Cx = d\}$.

♡ **ellipsoid** $\{x \mid (x - x_0)^T A (x - x_0) \leq 1\}$, (A : semidefinite)

♡ **probability simplex** $\{x \mid x \geq 0, \sum_i x_i = 1\}$

- Define the relative interior

- Define the asymptotic cone

Convex functions

Def. Function $f : I \rightarrow \mathbb{R}$ on interval I called **midpoint convex** if

$$f\left(\frac{x+y}{2}\right) \leq \frac{f(x)+f(y)}{2}, \quad \text{whenever } x, y \in I.$$

Read: f of AM is less than or equal to AM of f .

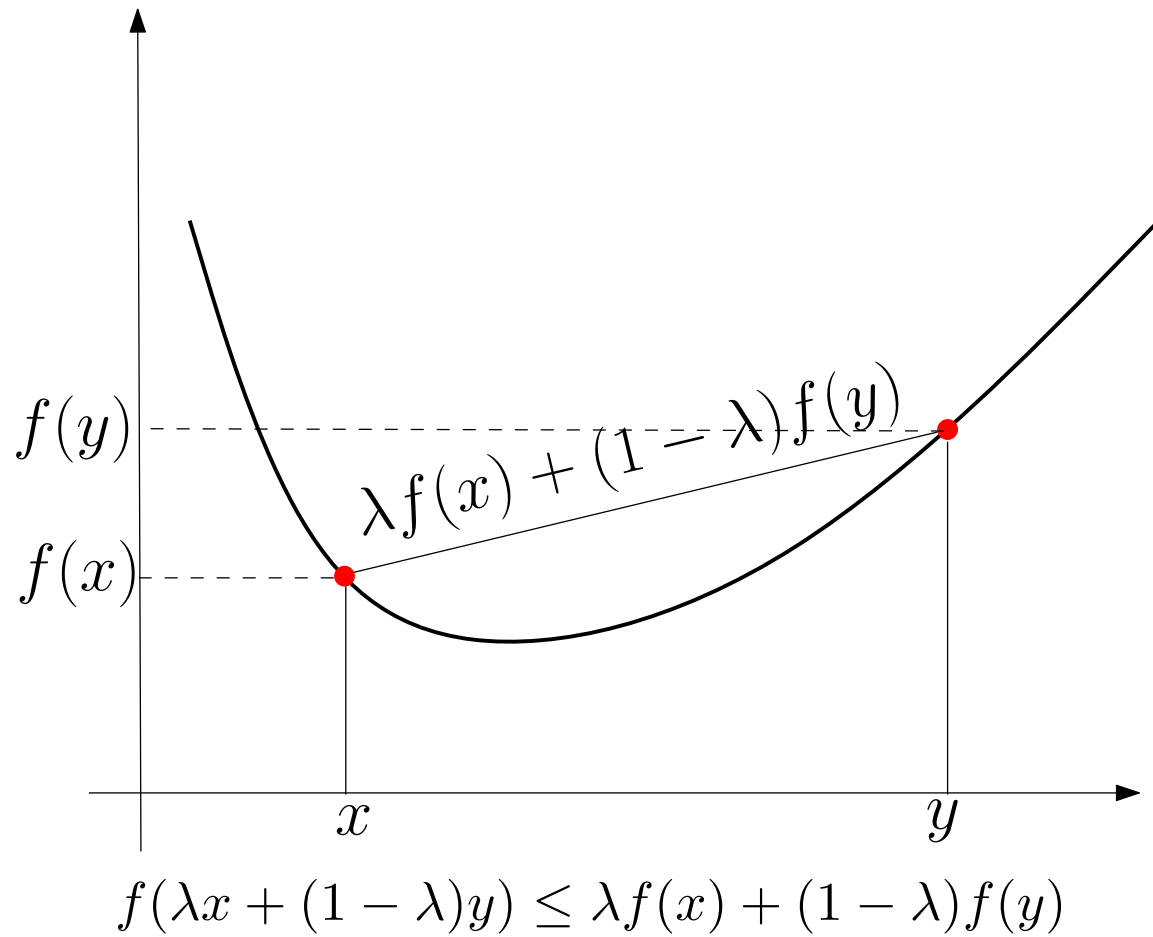
Def. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is called **convex** if its domain $\text{dom}(f)$ is a convex set and for any $x, y \in \text{dom}(f)$ and $\theta \in [0,1]$

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y).$$

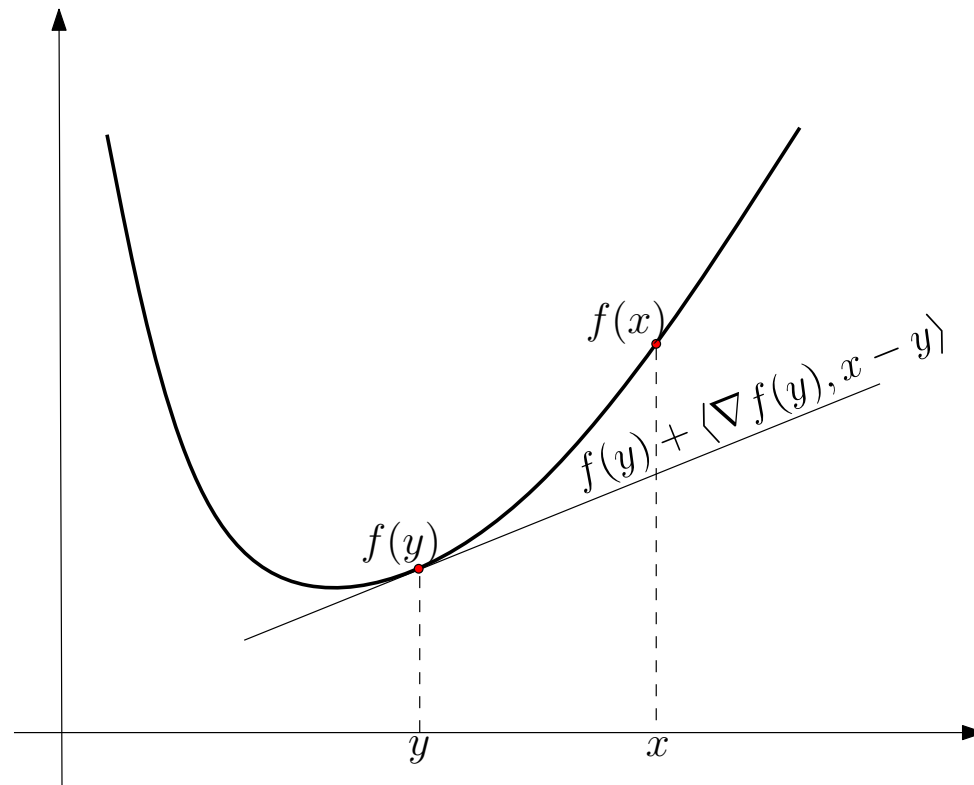
Theorem (J.L.W.V. Jensen). Let $f : I \rightarrow \mathbb{R}$ be continuous. Then, f is convex *if and only if* it is midpoint convex.

► Theorem extends to functions $f : \mathcal{X} \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. Very useful to checking convexity of a given function.

Convex functions

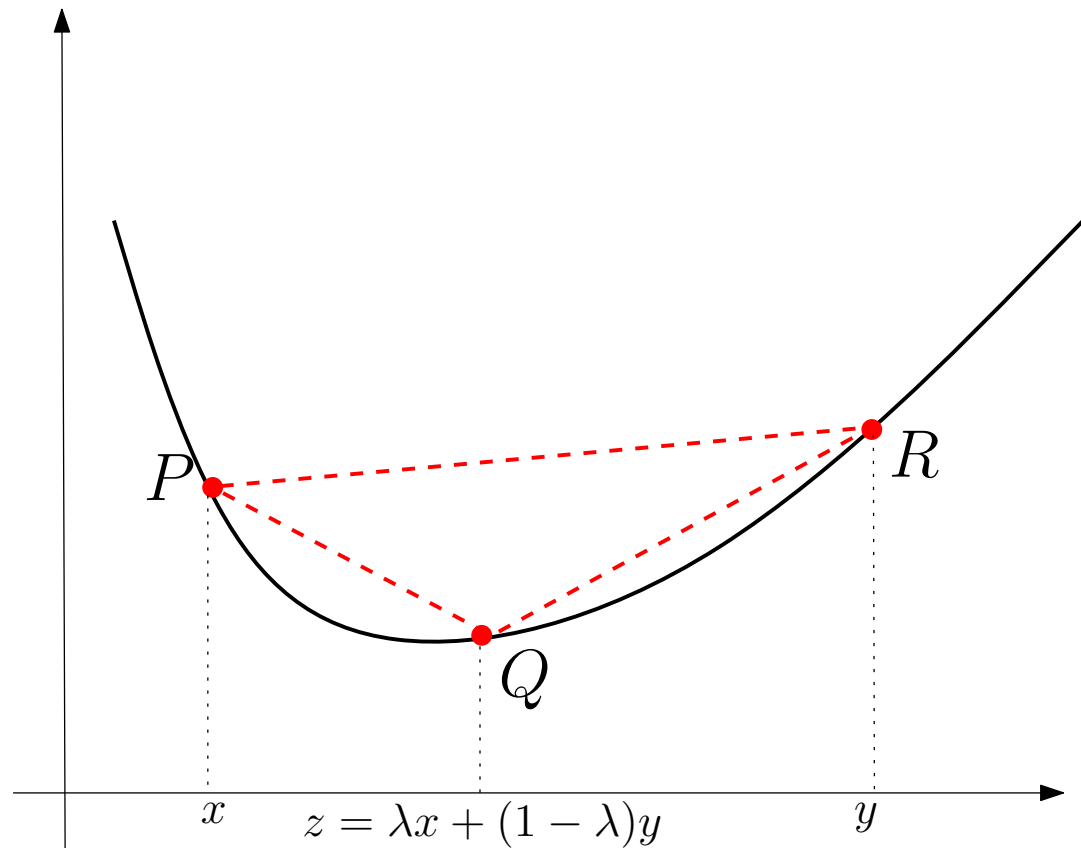


Convex functions



$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$$

Convex functions



$$\text{slope } PQ \leq \text{slope } PR \leq \text{slope } QR$$

Recognizing convex functions

- ♠ If f is continuous and midpoint convex, then it is convex.
- ♠ If f is differentiable, then f is convex *if and only if* $\text{dom } f$ is convex and $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$ for all $x, y \in \text{dom } f$.
- ♠ If f is twice differentiable, then f is convex *if and only if* $\text{dom } f$ is convex and $\nabla^2 f(x) \succeq 0$ at every $x \in \text{dom } f$.

Convex functions

- Linear: $f(\theta_1x + \theta_2y) = \theta_1f(x) + \theta_2f(y)$; θ_1, θ_2 unrestricted
- Concave: $f(\theta x + (1 - \theta)y) \geq \theta f(x) + (1 - \theta)f(y)$
- Strictly convex: If inequality is strict for $x \neq y$

Convex functions

Example The *pointwise maximum* of a family of convex functions is convex. That is, if $f(x; y)$ is a convex function of x for every y in some “index set” $\mathcal{Y}$, then

$$f(x) := \max_{y \in \mathcal{Y}} f(x; y)$$

is a convex function of x (set $\mathcal{Y}$ is arbitrary).

Example Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be convex. Let $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$. Prove that $g(x) = f(Ax + b)$ is convex.

Convex functions

Theorem Let $\mathcal{Y}$ be a nonempty convex set. Suppose $L(x, y)$ is convex in (x, y) , then,

$$f(x) := \inf_{y \in \mathcal{Y}} L(x, y)$$

is a convex function of x , provided $f(x) > -\infty$.

Proof. Let $u, v \in \text{dom } f$. Since $f(u) = \inf_y L(u, y)$, for each $\epsilon > 0$, there is a $y_1 \in \mathcal{Y}$, s.t. $f(u) + \frac{\epsilon}{2}$ is **not** the infimum. Thus, $L(u, y_1) \leq f(u) + \frac{\epsilon}{2}$. Similarly, there is $y_2 \in \mathcal{Y}$, such that $L(v, y_2) \leq f(v) + \frac{\epsilon}{2}$.

Now we prove that $f(\lambda u + (1 - \lambda)v) \leq \lambda f(u) + (1 - \lambda)f(v)$ directly.

$$\begin{aligned} f(\lambda u + (1 - \lambda)v) &= \inf_{y \in \mathcal{Y}} L(\lambda u + (1 - \lambda)v, y) \\ &\leq L(\lambda u + (1 - \lambda)v, \lambda y_1 + (1 - \lambda)y_2) \\ &\leq \lambda L(u, y_1) + (1 - \lambda)L(v, y_2) \\ &\leq \lambda f(u) + (1 - \lambda)f(v) + \epsilon. \end{aligned}$$

Since $\epsilon > 0$ is arbitrary, claim follows.

Example: Schur complement

Let A, B, C be matrices such that $C \succ 0$, and let

$$Z := \begin{bmatrix} A & B \\ B^T & C \end{bmatrix} \succeq 0,$$

then the **Schur complement** $A - BC^{-1}B^T \succeq 0$.

Proof. $L(x, y) = [x, y]^T Z [x, y]$ is convex in (x, y) since $Z \succeq 0$

Observe that $f(x) = \inf_y L(x, y) = x^T (A - BC^{-1}B^T)x$ is convex.

Exercise:

(We skipped ahead and solved $\nabla_y L(x, y) = 0$ to minimize).

Recognizing convex functions

- ♠ If f is continuous and midpoint convex, then it is convex.
- ♠ If f is differentiable, then f is convex *if and only if* $\text{dom } f$ is convex and $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$ for all $x, y \in \text{dom } f$.
- ♠ If f is twice differentiable, then f is convex *if and only if* $\text{dom } f$ is convex and $\nabla^2 f(x) \succeq 0$ at every $x \in \text{dom } f$.
- ♠ By showing f to be a pointwise max of convex functions
- ♠ By showing $f : \text{dom}(f) \rightarrow \mathbb{R}$ is convex *if and only if* its restriction to **any line** that intersects $\text{dom}(f)$ is convex. That is, for any $x \in \text{dom}(f)$ and any v , the function $g(t) = f(x + tv)$ is convex (on its domain $\{t \mid x + tv \in \text{dom}(f)\}$).

Operations preserving convexity

Pointwise maximum: $f(x) = \sup_{y \in \mathcal{Y}} f(y; x)$

Conic combination: Let $a_1, \dots, a_n \geq 0$; let $f_1, \dots, f_n$ be convex functions. Then, $f(x) := \sum_i a_i f_i(x)$ is convex.

Remark: The set of all convex functions is a *convex cone*.

Affine composition: $f(x) := g(Ax + b)$, where g is convex.

Operations preserving convexity

Theorem Let $f : I_1 \rightarrow \mathbb{R}$ and $g : I_2 \rightarrow \mathbb{R}$, where $\text{range}(f) \subseteq I_2$. If f and g are convex, and g is **increasing**, then $g \circ f$ is convex on I_1

Proof. Let $x, y \in I_1$, and let $\lambda \in (0, 1)$.

$$\begin{aligned} f(\lambda x + (1 - \lambda)y) &\leq \lambda f(x) + (1 - \lambda)f(y) \\ g(f(\lambda x + (1 - \lambda)y)) &\leq g(\lambda f(x) + (1 - \lambda)f(y)) \\ &\leq \lambda g(f(x)) + (1 - \lambda)g(f(y)). \end{aligned}$$

Quadratic

Let $f(x) = x^T A x + b^T x + c$, where $A \succeq 0$, $b \in \mathbb{R}^n$, and $c \in \mathbb{R}$.

What is: $\nabla^2 f(x)$?

$\nabla f(x) = 2Ax + b$, $\nabla^2 f(x) = A \succeq 0$, hence f is convex.

Indicator

Let $\mathbb{I}_{\mathcal{X}}$ be the *indicator function* for $\mathcal{X}$ defined as:

$$\mathbb{I}_{\mathcal{X}}(x) := \begin{cases} 0 & \text{if } x \in \mathcal{X}, \\ \infty & \text{otherwise.} \end{cases}$$

Note: $\mathbb{I}_{\mathcal{X}}(x)$ is convex **if and only if** $\mathcal{X}$ is convex.

Distance to a set

Example Let $\mathcal{Y}$ be a convex set. Let $x \in \mathbb{R}^n$ be some point. The distance of x to the set $\mathcal{Y}$ is defined as

$$\text{dist}(x, \mathcal{Y}) := \inf_{y \in \mathcal{Y}} \|x - y\|.$$

Because $\|x - y\|$ is jointly convex in (x, y) , the function $\text{dist}(x, \mathcal{Y})$ is a convex function of x .

Norms

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function that satisfies

- 1 $f(x) \geq 0$, and $f(x) = 0$ if and only if $x = 0$ (**definiteness**)
- 2 $f(\lambda x) = |\lambda|f(x)$ for any $\lambda \in \mathbb{R}$ (**positive homogeneity**)
- 3 $f(x + y) \leq f(x) + f(y)$ (**subadditivity**)

Such a function is called a **norm**. We usually denote norms by $\|x\|$.

Theorem Norms are convex.

Proof. Immediate from subadditivity and positive homogeneity.

Dual norms

Def. Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. Its **dual norm** is

$$\|u\|_* := \sup \{ u^T x \mid \|x\| \leq 1 \}.$$

Exercise: Verify that $\|u\|_*$ is a norm.

Exercise: Let $1/p + 1/q = 1$, where $p, q \geq 1$. Show that $\|\cdot\|_q$ is dual to $\|\cdot\|_p$. In particular, the ℓ_2 -norm is self-dual.

Fenchel conjugate

Def. The **Fenchel conjugate** of a function f is

$$f^*(z) := \sup_{x \in \text{dom } f} x^T z - f(x).$$

Note: f^* is pointwise (over x) sup of linear functions of z . Hence, it is always convex (regardless of f).

Example $+\infty$ and $-\infty$ conjugate to each other.

Example Let $f(x) = \|x\|$. We have $f^*(z) = \mathbb{I}_{\|\cdot\|_* \leq 1}(z)$. That is, conjugate of norm is the indicator function of dual norm ball.

$f^*(z) = \sup_x z^T x - \|x\|$. If $\|z\|_* > 1$, then by definition of the dual norm, there is u s.t. $\|u\| \leq 1$ and $u^T z > 1$. Now select $x = \alpha u$ and let $\alpha \rightarrow \infty$. Then, $z^T x - \|x\| = \alpha(z^T u - \|u\|) \rightarrow \infty$. If $\|z\|_* \leq 1$, then $z^T x \leq \|x\| \|z\|_*$, which implies the sup must be zero.

Fenchel conjugate

Example $f(x) = ax + b$; then,

$$\begin{aligned} f^*(z) &= \sup_x zx - (ax + b) \\ &= \infty, \quad \text{if } (z - a) \neq 0. \end{aligned}$$

Thus, $\text{dom } f^* = \{a\}$, and $f^*(a) = -b$.

Example Let $a \geq 0$, and set $f(x) = -\sqrt{a^2 - x^2}$ if $|x| \leq a$, and $+\infty$ otherwise. Then, $f^*(z) = a\sqrt{1 + z^2}$.

Example $f(x) = \frac{1}{2}x^T Ax$, where $A \succ 0$. Then, $f^*(z) = \frac{1}{2}z^T A^{-1}z$.

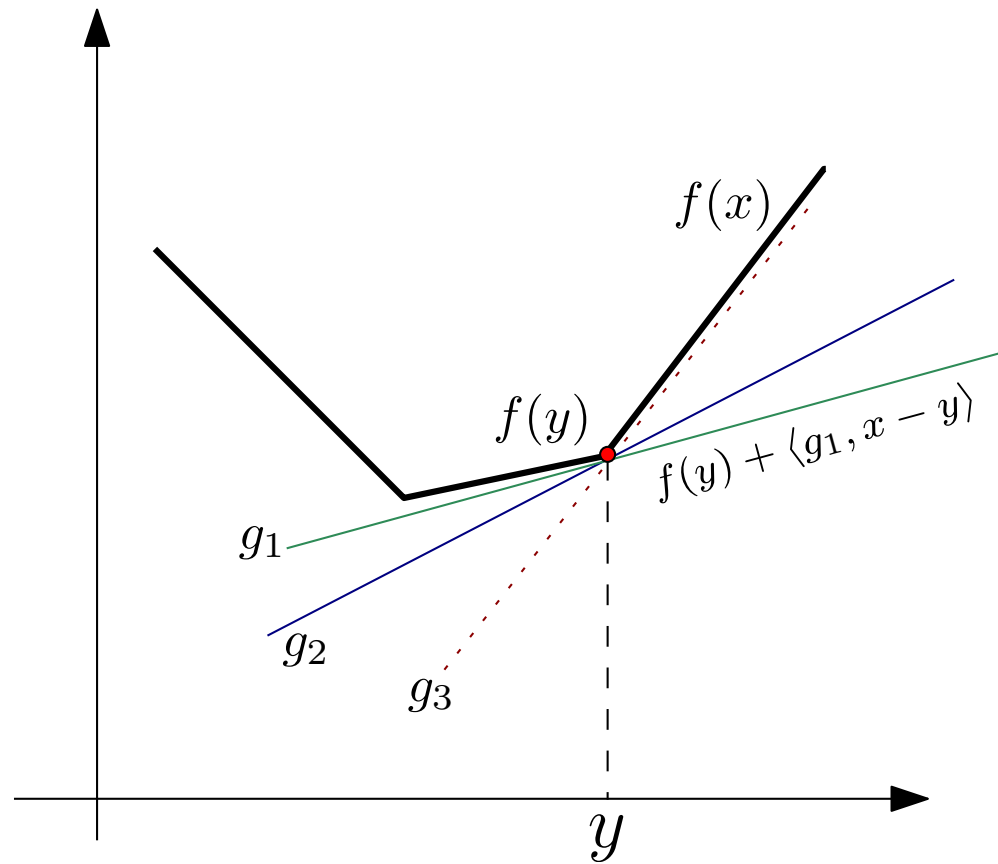
Example $f(x) = \max(0, 1 - x)$. Now $f^*(z) = \sup_x zx - \max(0, 1 - x)$. Note that $\text{dom } f^*$ is $[-1, 0]$ (else sup is unbounded); within this domain, $f^*(z) = z$.

Other forms of convexity

- ♣ **Log-convex:** $\log f$ is convex; $\log\text{-cvx} \implies \text{cvx}$;
- ♣ **Log-concavity:** $\log f$ concave; **not** closed under addition!
- ♣ **Exponentially convex:** $[f(x_i + x_j)] \succeq 0$, for $x_1, \dots, x_n$
- ♣ **Operator convex:** $f(\lambda X + (1 - \lambda)Y) \preceq \lambda f(X) + (1 - \lambda)f(Y)$
- ♣ **Quasiconvex:** $f(\lambda x + (1 - \lambda)y) \leq \max\{f(x), f(y)\}$
- ♣ **Pseudoconvex:** $\langle \nabla f(y), x - y \rangle \geq 0 \implies f(x) \geq f(y)$

Subdifferentials

Subgradients



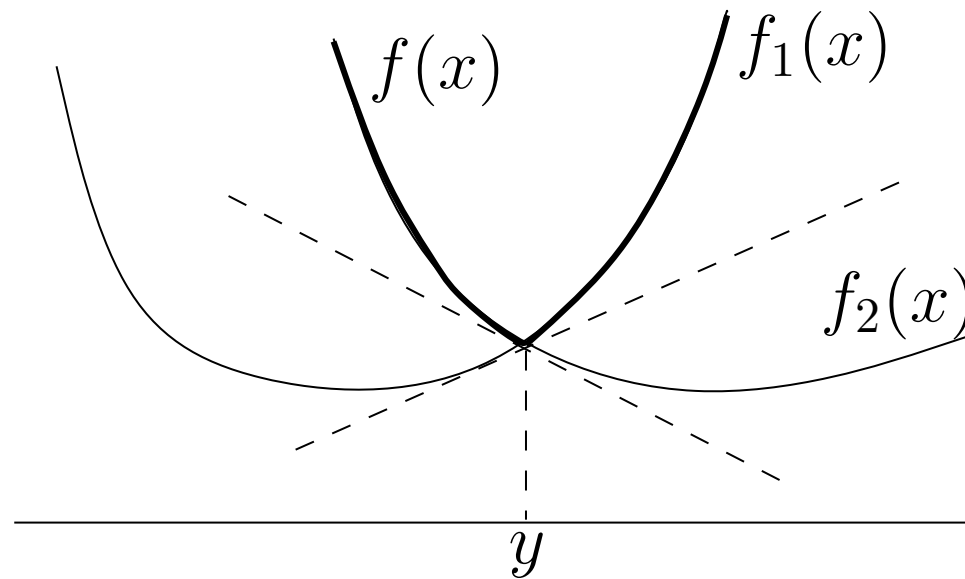
g_1, g_2, g_3 are subgradients at y

Subgradients – basic facts

- ▶ f is convex, differentiable: $\nabla f(y)$ the **unique** subgradient at y
- ▶ A vector g is a subgradient at a point y if and only if $f(y) + \langle g, x - y \rangle$ is **globally** smaller than $f(x)$.
- ▶ Usually, **one** subgradient costs approx. as much as $f(x)$
- ▶ Determining all subgradients at a given point — **difficult**.
- ▶ Subgradient calculus—a great achievement in convex analysis
- ▶ Without convexity, things become wild! — advanced course

Subgradients – example

$f(x) := \max(f_1(x), f_2(x))$; both f_1, f_2 convex, differentiable



- ★ $f_1(x) > f_2(x)$: unique subgradient of f is $f'_1(x)$
- ★ $f_1(x) < f_2(x)$: unique subgradient of f is $f'_2(x)$
- ★ $f_1(y) = f_2(y)$: subgradients, the segment $[f'_1(y), f'_2(y)]$
(imagine all supporting lines turning about point y)

Subgradients

Def. A vector $g \in \mathbb{R}^n$ is called a **subgradient** at a point y , if **for all** $x \in \text{dom } f$, it holds that

$$f(x) \geq f(y) + \langle g, x - y \rangle$$

Subdifferential

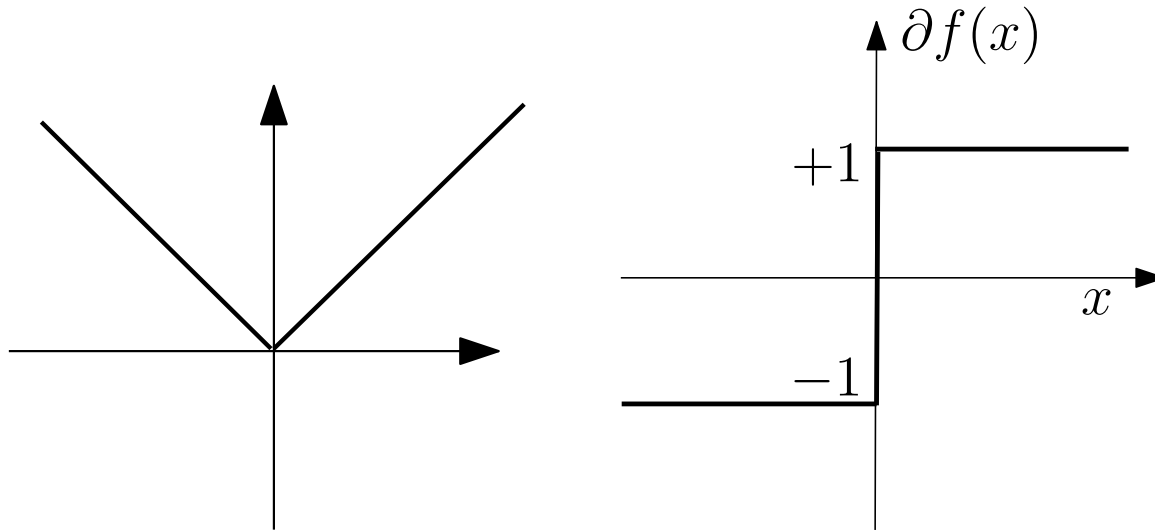
Def. The set of all subgradients at y denoted by $\partial f(y)$. This set is called **subdifferential** of f at y

If f is convex, $\partial f(x)$ is nice:

- ♣ If $x \in$ relative interior of $\text{dom } f$, then $\partial f(x)$ nonempty
- ♣ If f differentiable at x , then $\partial f(x) = \{\nabla f(x)\}$
- ♣ If $\partial f(x) = \{g\}$, then f is differentiable and $g = \nabla f(x)$

Subdifferential – example

$$f(x) = |x|$$



$$\partial|x| = \begin{cases} -1 & x < 0, \\ +1 & x > 0, \\ [-1, 1] & x = 0. \end{cases}$$

More examples

Example $f(x) = \|x\|_2$. Then,

$$\partial f(x) := \begin{cases} \|x\|_2^{-1}x & x \neq 0, \\ \{z \mid \|z\|_2 \leq 1\} & x = 0. \end{cases}$$

Proof.

$$\begin{aligned} \|z\|_2 &\geq \|x\|_2 + \langle g, z - x \rangle \\ \|z\|_2 &\geq \langle g, z \rangle \\ \implies \|g\|_2 &\leq 1. \end{aligned}$$

More examples

Example A convex function need not be subdifferentiable everywhere.

Let

$$f(x) := \begin{cases} -(1 - \|x\|_2^2)^{1/2} & \text{if } \|x\|_2 \leq 1, \\ +\infty & \text{otherwise.} \end{cases}$$

f diff. for all x with $\|x\|_2 < 1$, but $\partial f(x) = \emptyset$ whenever $\|x\|_2 \geq 1$.

Recall basic calculus

If f and k are differentiable, we know that

- **Addition:** $\nabla(f + k)(x) = \nabla f(x) + \nabla k(x)$
- **Scaling:** $\nabla(\alpha f(x)) = \alpha \nabla f(x)$

Chain rule

If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and $k : \mathbb{R}^m \rightarrow \mathbb{R}^p$. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}^p$ be the composition $h(x) = (k \circ f)(x) = k(f(x))$. Then,

$$Dh(x) = Dk(f(x))Df(x).$$

Example If $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $k : \mathbb{R} \rightarrow \mathbb{R}$, then using the fact that $\nabla h(x) = [Dh(x)]^T$, we obtain

$$\nabla h(x) = k'(f(x))\nabla f(x).$$

Subgradient calculus

- ♠ Finding **one** subgradient within $\partial f(x)$
- ♠ Determining entire subdifferential $\partial f(x)$ at a point x
- ♠ Do we have the chain rule?
- ♠ Usually not easy!

Subgradient calculus

§ If f is differentiable, $\partial f(x) = \{\nabla f(x)\}$

§ **Scaling** $\alpha > 0$, $\partial(\alpha f)(x) = \alpha \partial f(x) = \{\alpha g \mid g \in \partial f(x)\}$

§ **Addition***: $\partial(f + k)(x) = \partial f(x) + \partial k(x)$ (set addition)

§ **Chain rule***: Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, $f : \mathbb{R}^m \rightarrow \mathbb{R}$, and $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be given by $h(x) = f(Ax + b)$. Then,

$$\partial h(x) = A^T \partial f(Ax + b).$$

§ **Chain rule***: $h(x) = f \circ k$, where $k : X \rightarrow Y$ is diff.

$$\partial h(x) = \partial f(k(x)) \circ Dk(x) = [Dk(x)]^T \partial f(k(x))$$

§ **Max function***: If $f(x) := \max_{1 \leq i \leq m} f_i(x)$, then

$$\partial f(x) = \text{conv} \bigcup \{ \partial f_i(x) \mid f_i(x) = f(x) \},$$

convex hull over subdifferentials of “active” functions at x

§ **Conjugation**: $z \in \partial f(x)$ if and only if $x \in \partial f^*(z)$

Examples

It can happen that $\partial(f_1 + f_2) \neq \partial f_1 + \partial f_2$

Example Define f_1 and f_2 by

$$f_1(x) := \begin{cases} -2\sqrt{x} & \text{if } x \geq 0, \\ +\infty & \text{if } x < 0, \end{cases} \quad \text{and} \quad f_2(x) := \begin{cases} +\infty & \text{if } x > 0, \\ -2\sqrt{-x} & \text{if } x \leq 0. \end{cases}$$

Then, $f = \max\{f_1, f_2\} = \mathbb{I}_0$, whereby $\partial f(0) = \mathbb{R}$

But $\partial f_1(0) = \partial f_2(0) = \emptyset$.

However, $\partial f_1(x) + \partial f_2(x) \subset \partial(f_1 + f_2)(x)$ always holds.

Examples

Example $f(x) = \|x\|_\infty$. Then,

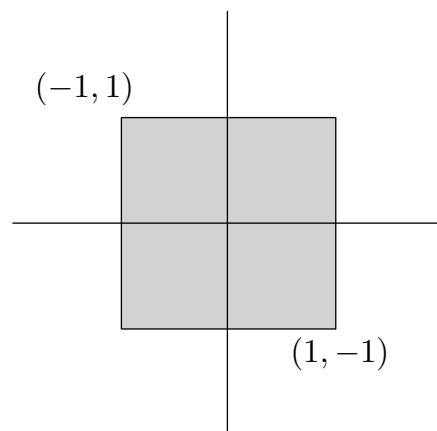
$$\partial f(0) = \text{conv} \{ \pm e_1, \dots, \pm e_n \},$$

where e_i is i -th canonical basis vector (column of identity matrix).

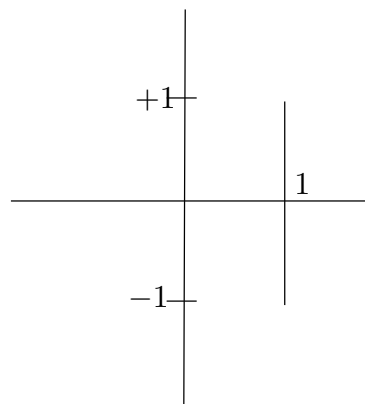
To prove, notice that $f(x) = \max_{1 \leq i \leq n} \{ |e_i^T x| \}$.

Example

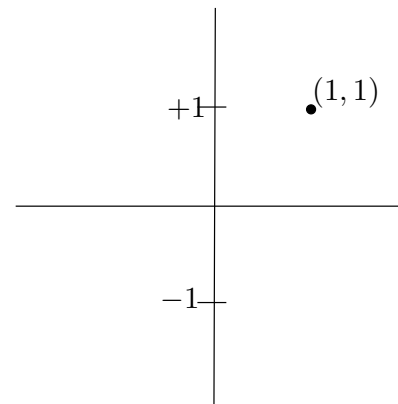
Example Let $f(x) = \max \{s^T x \mid s_i \in \{-1, 1\}\}$ (2^n members)



∂f at $x = (0,0)$



∂f at $x = (1,0)$



∂f at $x = (1,1)$

Subgradient for pointwise sup

$$f(x) := \sup_{y \in \mathcal{Y}} h(x, y)$$

Getting $\partial f(x)$ is complicated!

Simple way to obtain some $g \in \partial f(x)$:

- ▶ Pick **any** y^* for which $h(x, y^*) = f(x)$
- ▶ Pick **any** subgradient $g \in \partial h(x, y^*)$
- ▶ This $g \in \partial f(x)$

$$h(z, y^*) \geq h(x, y^*) + g^T(z - x)$$

$$h(z, y^*) \geq f(x) + g^T(z - x)$$

$$f(z) \geq h(z, y^*) \quad (\text{because of sup})$$

$$f(z) \geq f(x) + g^T(z - x).$$

Example

Suppose $a_i \in \mathbb{R}^n$ and $b_i \in \mathbb{R}$. And

$$f(x) := \max_{1 \leq i \leq n} (a_i^T x + b_i).$$

This f a max (in fact, over a finite number of terms)

- ▶ Suppose $f(x) = a_k^T x + b_k$ for some index k
- ▶ Here $f(x; y) = f_k(x) = a_k^T x + b_k$, and $\partial f_k(x) = \{\nabla f_k(x)\}$
- ▶ Hence, $a_k \in \partial f(x)$ works!

Subgradient of expectation

Suppose $f = \mathbf{E}f(x, u)$, where f is convex in x for each u (an r.v.)

$$f(x) := \int f(x, u)p(u)du$$

- ▶ For each u choose **any** $g(x, u) \in \partial_x f(x, u)$
- ▶ Then, $g = \int g(x, u)p(u)du = \mathbf{E}g(x, u) \in \partial f(x)$

Subgradient of composition

Suppose $h : \mathbb{R}^n \rightarrow \mathbb{R}$ cvx and **nondecreasing**; each f_i cvx

$$f(x) := h(f_1(x), f_2(x), \dots, f_n(x)).$$

To find a vector $g \in \partial f(x)$, we may:

- ▶ For $i = 1$ to n , compute $g_i \in \partial f_i(x)$
- ▶ Compute $u \in \partial h(f_1(x), \dots, f_n(x))$
- ▶ Set $g = u_1 g_1 + u_2 g_2 + \dots + u_n g_n$; this $g \in \partial f(x)$
- ▶ Compare with $\nabla f(x) = J \nabla h(x)$, where J matrix of $\nabla f_i(x)$

Exercise: Verify $g \in \partial f(x)$ by showing $f(z) \geq f(x) + g^T(z - x)$